Experiment 1: Thevenin's Theorem

OBJECTIVE:

1. To obtain Thevenin equivalent circuit from a complex circuit with independent source.
2. To perform comparison analysis between the complex circuit and the Thevenin equivalent circuits with independent source.
3. To show that Thevenin theorem are the simplest method to analyze load variation in a complex circuits.
4. To obtain Thevenin equivalent circuit with independent source and dependent source.

INTRODUCTION:

Thevenin's Theorem

Its states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source Vth in series with a resistor Rth, where Vth is the open-circuit voltage at the terminals and Rth is the input or equivalent resistance at the terminals when the independent sources are turned off.

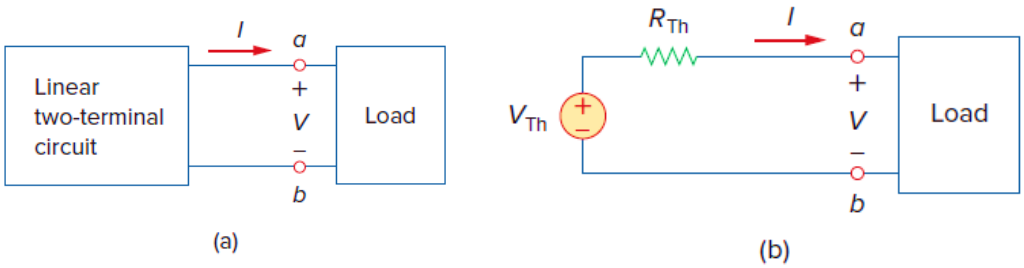
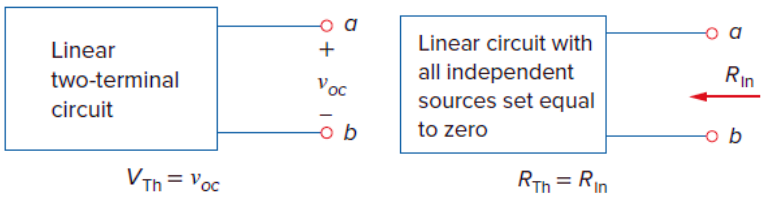


Figure 1: Thevenin's Theorem



(a) (b)

Figure 2: (a) Vth is the Open-circuit voltage across the terminals, (b) Rth is the input resistance at the terminals when the independent sources are turn off.

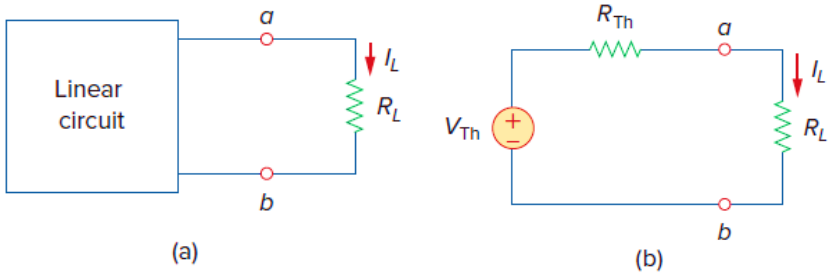


Figure 3: A circuit with a load: (a) original circuit, (b) Thevenin equivalent.

Formula:

$$Load\ Current, I_L = \frac{V_{th}}{R_{th} + R_L}$$

EQUIPMENT AND COMPONENT

1. Resistor: 5 Ω, 10 Ω, 12 Ω, 60 Ω, 100 Ω and 150 Ω
2. Load resistor: 0 Ω, 10 Ω, 20 Ω, 30 Ω and 50 Ω
3. DC power supply, 1V, 10V and 18V.
4. DC current source, 1A
5. Current Controlled Current Source, CCCS , Current gain = 2
6. Multimeter

PROCEDURE

Part A: Load current measurement in complex circuit with independence source



Figure 4: Multisim 14.0 software icon

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM106_Experiment 1_YOUR ID_YOURNAME > Save.
3. Follow Step 4 until Step 19 to construct your complex circuit as shown in figure 5.

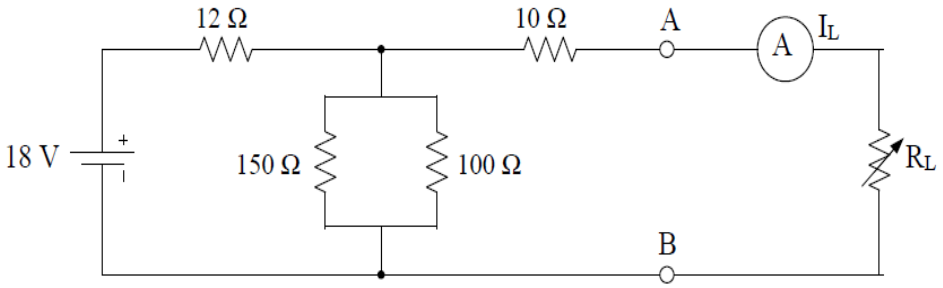


Figure 5: Complex circuit with independent source

4. Click Place > Component > Group > Sources > POWER_SOURCES > DC_POWER > OK. Place it on the screen.
5. Set your DC_POWER to 18 V. Double click on DC_POWER icon > Value > Voltage > 18V > OK.
6. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.
7. Set your RESISTOR value to 12 Ω. Double click on RESISTOR > Value > Resistance > 12 Ω > OK.
8. To rotate the RESISTOR > Click RESISTOR > Ctrl + R **or** Right click > Rotate 90^0 Clockwise/Counterclockwise.
9. Repeat step 6 until step 8 to set complete RESISTOR in the circuit.
10. Click Place > Component > Group > Sources > POWER_SOURCES > GROUND > OK. Place it on the screen.
11. Connect all component as shown in Figure 5.
12. To label load resistor. Double click on RESISTOR > Label > RL > OK.

13. Set your RL to 0Ω. Double click on RESISTOR > Value > Resistance > 0 Ω > OK.
14. To label the terminal A-B. Click Place > Text > A.
15. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in series connection with the circuit for load current measurement.

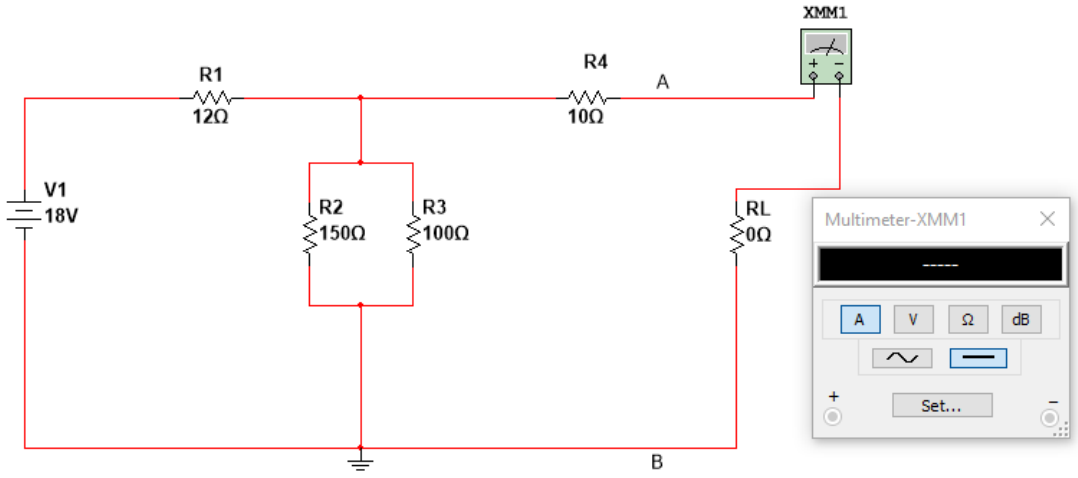


Figure 6: Complex circuit with Ammeter

16. Double click Multimeter > Choose A for current measurement > Straight line for DC current.

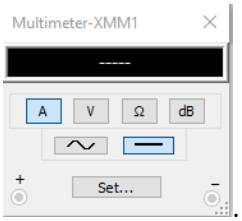


Figure 7: Ammeter setting

17. Run the simulation

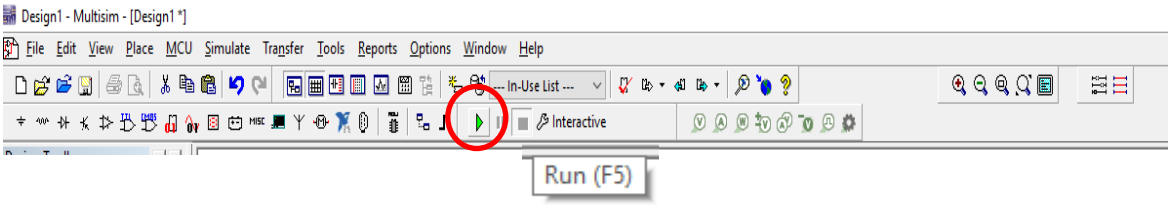


Figure 8: Run button/Icon

18. Measure and record the ammeter reading for load resistor, R_L = 0Ω in Table 1.
19. Repeat step 13 to change your load resistor to 10Ω, 20Ω, 30Ω and 50Ω and run again your circuit simulation. Records your load current measurement in Table 1.

Part B: Thevenin Theorem with independence source

1. By using the same circuit in Part A, open the terminals A and B by using the provided meters.
2. Measure the Thevenin equivalent resistance, R_{th} at terminals A and B. Turn off DC power source by make it as short circuit.

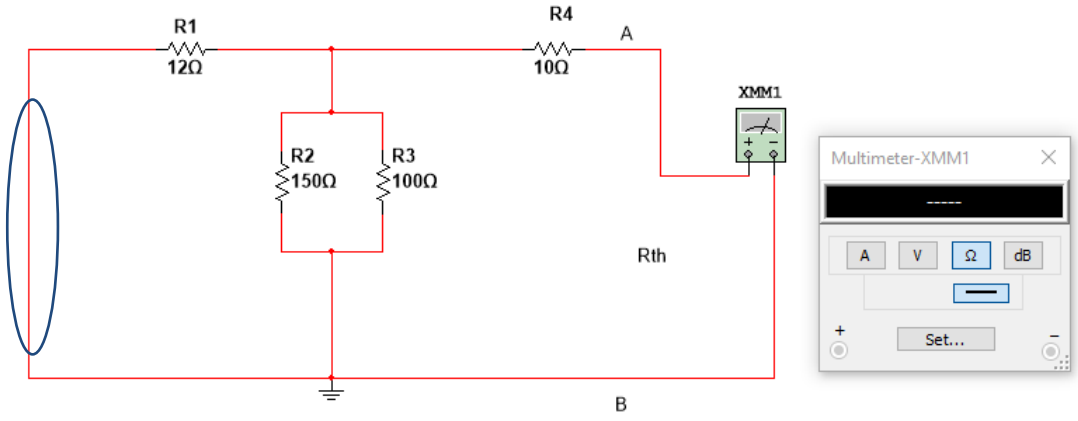


Figure 9: Complex circuit with off condition of independent source to measure Rth

3. Double click Multimeter > Choose **Ω** for resistance measurement. Run the simulation circuit and record your measurement in Table 2.
4. Measure the open circuit voltage, V_{oc} or Thevenin Voltage, V_{th} at terminal A and B by replacing back the DC power supply.

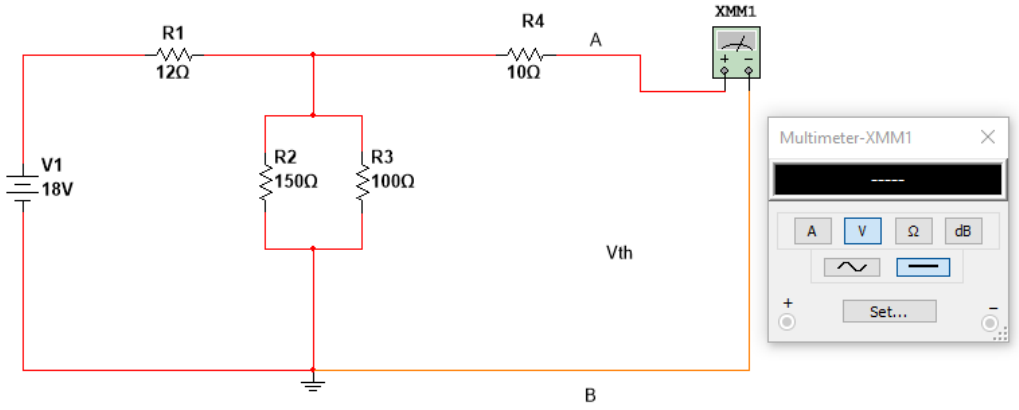


Figure 10: Complex circuit with open circuit condition at output voltage

5. Repeat step 5 in Part A.
6. Double click Multimeter > Choose V for voltage measurement > Click straight line for Dc voltage measurement. Run the simulation circuit and record your measurement in Table 2.

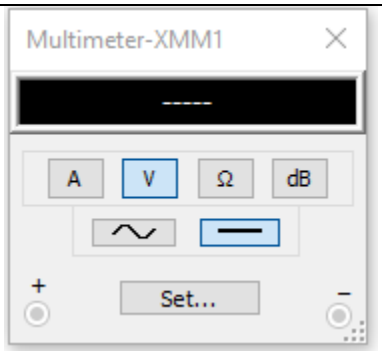


Figure 11: DC Voltmeter setting

7. By using the measured value of Thevenin equivalent resistance, R_{th} and Thevenin Voltage, V_{th}, connect the Thevenin circuit as shown in figure 12 below. Measured the load current, I_L for load resistor, R_L = 0Ω and records the value in Table 3.

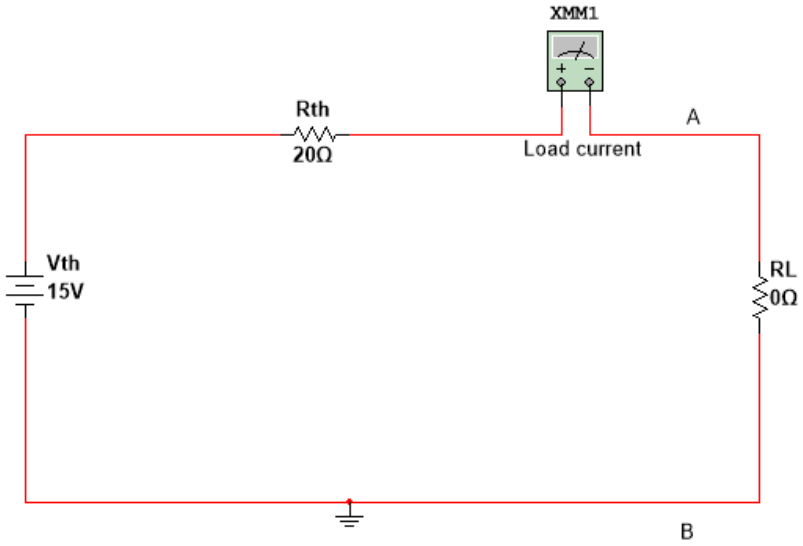


Figure 12: Thevenin circuit with load resistance, R_L

8. Repeat the step 7 for load resistor, R_L = 10Ω, 20Ω, 30Ω and 50Ω. Records the load current measurement value in Table 3.

Part C: Thevenin Theorem circuit with independent and dependent source

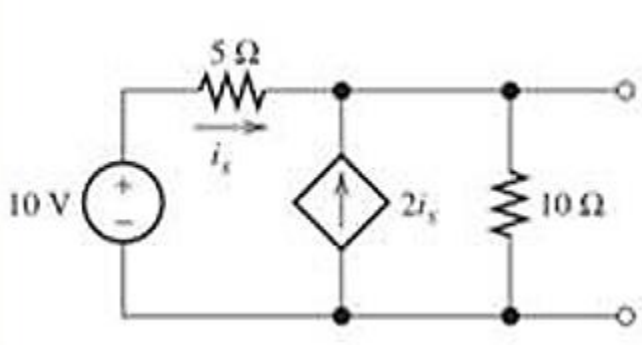


Figure 13: complex circuit with independent and dependent source

1. Follow Step 2 until Step 19 to construct and find your Thevenin circuit with independent and dependent source.
2. Find open circuit voltage, Voc or Thevenin equivalent voltage, Vth. Click Place > Component > Group > Sources > POWER_SOURCES > DC_POWER > OK. Place it on the screen.
3. Set your DC_POWER to 10 V. Double click on DC_POWER icon > Value > Voltage > 10V > OK.
4. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.
5. Set your RESISTOR value to 5Ω. Double click on RESISTOR > Value > Resistance > 5 Ω > OK.
6. Repeat step 4 until step 5 to set complete RESISTOR in the circuit.
7. Set your dependent source gain to 2 A/A. Click Place > Component > Group > Sources > CONTROLLED_CURRENT_SOURCES>CURRENT_CONTROLLED_CURRENT_SOURCE> OK. Place it on the screen.
8. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in parallel connection to measure open circuit voltage, Voc.
9. Connect all component as shown in Figure 14.
10. Run the simulation. Measured the Thevenin equivalent voltage, Vth and record your answer in Table 4.1.

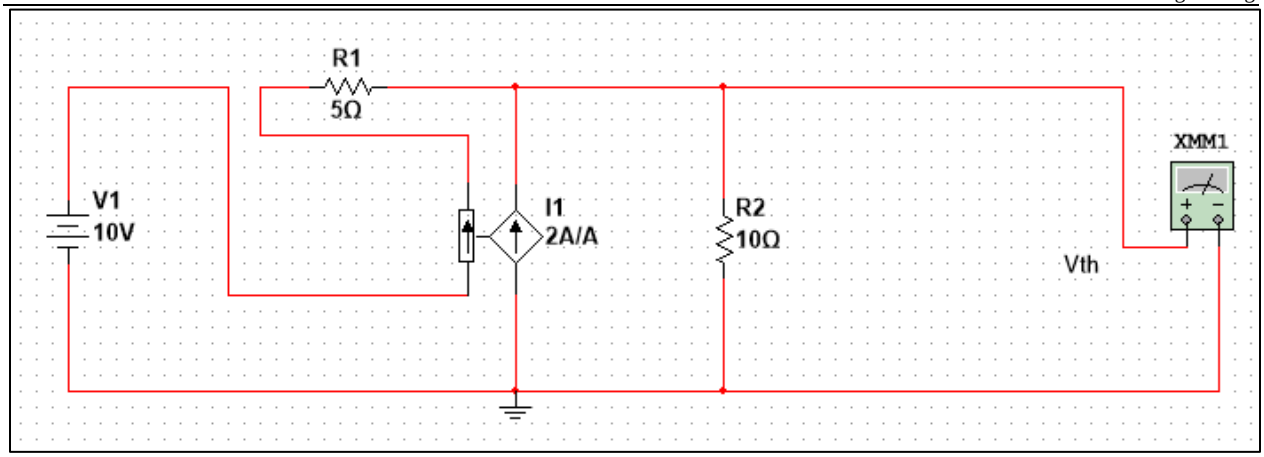


Figure 14: Complex circuit with independent and dependent source

11. There are three (3) method to find Thevenin equivalent resistance, Rth with independent and dependent source.
12. **Method 1**.By add on voltage source, 1V at terminals A-B. Turned off the independent source by short circuit. Connect the circuit diagram as shown in figure 15.

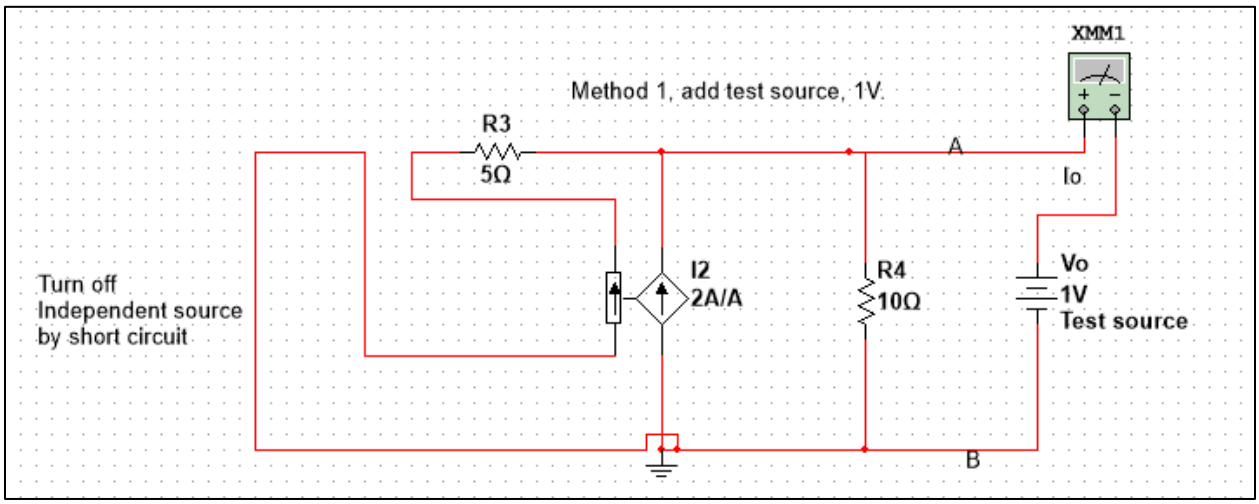


Figure 15: Method 1. Complex circuit with independent and dependent source to find Thevenin equivalent resistance, Rth with 1V voltage source.

13. Run the simulation. Measured the current, Io and calculate the Thevenin equivalent resistance, Rth using Ohm's law, and record your answer in Table 4.2.

$$R_{th} = \frac{V_O}{I_O}$$

14. **Method 2**. By using the same circuit diagram as shown in figure 15 by replacing the voltage source, 1V with DC current source, 1A.
15. Click Place > Component > Group > Sources >SIGNAL_CURRENT_SOURCES>DC_CURRENT > OK. Place it on the screen. As shown in figure 16.

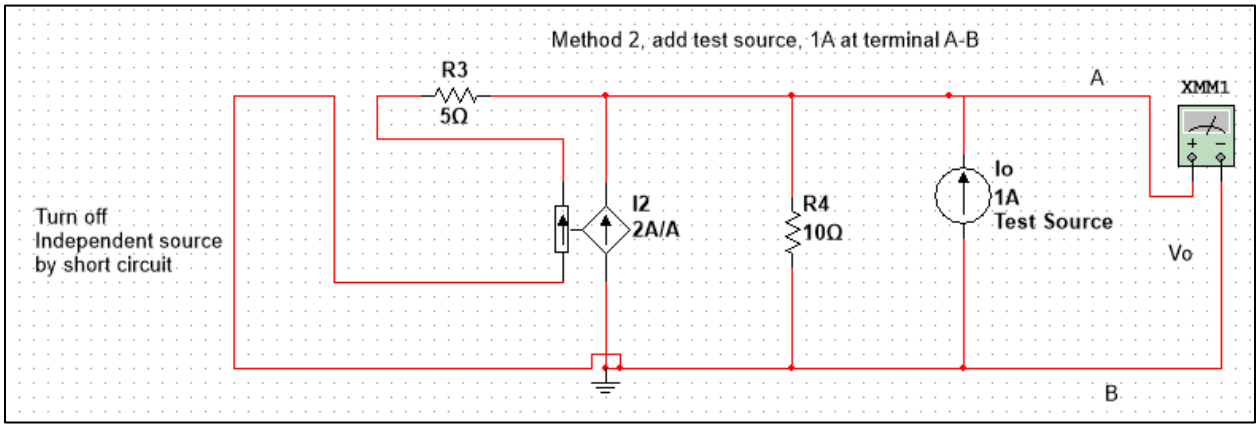


Figure 16: Complex circuit with independent and dependent source to find Thevenin equivalent resistance, Rth with 1A current source.

16. Run the simulation. Measured the voltage, Vo and calculate the Thevenin equivalent resistance, Rth using Ohm's law, and record your answer in Table 4.2.

$$R_{th} = \frac{V_o}{I_o}$$

17. **Method 3**. Short circuit method. While using short circuit method, the independent source should not be disturb. By using the same circuit as in figure 14, find the short circuit current between the terminals A-B as shown in figure 17.

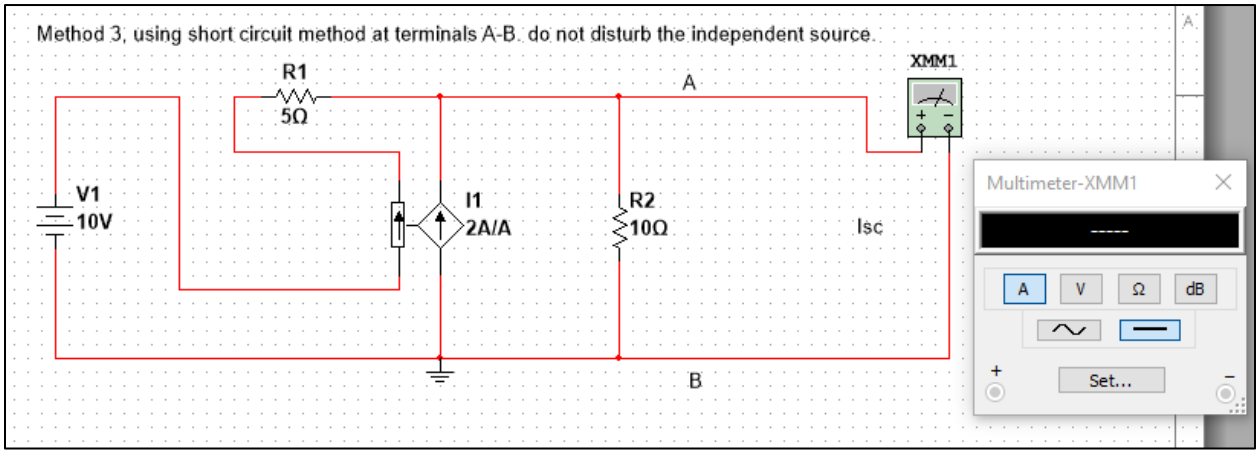


Figure 17: Complex circuit with independent and dependent source to find Thevenin equivalent resistance, Rth with short circuit current,Isc.

18. Records the short circuit current, Isc between terminals A-B in Table 4.2.
19. Run the simulation. Measured the short circuit current, Isc and calculate the Thevenin equivalent resistance, Rth using Ohm's law, and record your answer in Table 4.2.

$$R_{th} = \frac{V_{th}}{I_{sc}}$$

RESULT PART A

Part A: Load current measurement in complex circuit with independent source.

Table 1: Load current, I_L in complex circuit

Load Resistor, R_L (Ω)	Measured value I_L (mA)	Calculated value I_L (mA)
0		
10		
20		
30		
50		

RESULT PART B

Part B: Thevenin's Theorem with independence source.

Table 2: Thevenin theorem measurement with independent source

	Measured value	Calculated value
R_{TH} (Ω)		
V_{TH} (V)		

Table 3: Load current, I_L in Thevenin circuit with independent source

Load Resistor, R_L (Ω)	Measured value I_L (mA)	Calculated value I_L (mA)
0		
10		
20		
30		
50		

RESULT PART C

Part C: Thevenin Theorem circuit with independent and dependent source

Table 4.1: Thevenin equivalent voltage, Vth with independent and dependent source

Thevenin Equivalent Voltage, Vth	Measured value	Calculated value
V_{TH} (V)		

Table 4.2: Thevenin Equivalent resistor, Rth with independent and dependent source

Thevenin Equivalent Resistor, R_{TH}	Measured value	Calculated value
Method 1 (1V test source)		
Method 2 (1A test source)		
Method 3 (Short Circuit, Isc)		

DISCUSSION

The following points should be discuss in your report.

1. Analyze your result and verify the Thevenin theorem.
2. Discuss briefly the importance of Thevenin Theorem in circuit analysis.
3. Show your working calculation for Table 1, 2, 3, 4.1 and 4.2.